

3.8 Remembering Things

The chapter's capstone

Takes the raw file I/O from 3.7 and wraps it in functions, then uses them properly:

1. **Load** whatever friends are already saved
2. **Show** them
3. Let the user **add more**
4. **Save** the combined list back

The friends list grows a little more every time you run the program.

Wrapping file I/O in functions

```
std::vector<std::string> loadFriendsFromFile(const std::string& fileName) {  
    std::vector<std::string> friends;  
    std::ifstream inputFile(fileName);  
  
    std::string friendName;  
    while (std::getline(inputFile, friendName)) {  
        friends.push_back(friendName);  
    }  
  
    return friends;  
}
```

The other half

```
void saveFriendsToFile(const std::vector<std::string>& friends,
                      const std::string& fileName) {
    std::ofstream outputFile(fileName);
    for (const std::string& friendName : friends) {
        outputFile << friendName << "\n";
    }
    std::println("Saved {} friend(s) to {}", friends.size(), fileName);
}
```

Load first, THEN collect more

```
std::vector<std::string> friends = loadFriendsFromFile(fileName);

std::println("\nYou already have {} friend(s) saved:", friends.size());
for (const std::string& friendName : friends) {
    std::println(" - {}", friendName);
}

std::vector<std::string> newFriends = collectMoreFriendNames();
friends.insert(friends.end(), newFriends.begin(), newFriends.end());

saveFriendsToFile(friends, fileName);
```

Run it twice

First run - `friends.txt` doesn't exist yet:

```
You already have 0 friend(s) saved:
```

Second run - picks up right where you left off:

```
You already have 2 friend(s) saved:  
- Grace  
- Hopper
```

End of Chapter 3

First Steps